NEO Mathematics Learner Guide

Lesson 2 - Finding the Hypotenuse

Driving question: How can I recognise the hypotenuse and build a correct step-by-step solution for its length?

What this lesson is for

This lesson helps you practise one focused skill: finding the hypotenuse of a right-angled triangle. The hypotenuse is the side opposite the right angle. It may be on the left, on the right, at the base, or in a tilted triangle.

Stepping Stones

1. Find the right-angle marker first.
2. Trace to the side opposite the right angle. This side is the hypotenuse.
3. Use the correct lower-case letter for the side opposite the matching capital vertex: side opposite A is a, side opposite B is b, and side opposite C is c.
4. Write the theorem with the hypotenuse square on the left.
5. Square the two shorter side lengths.
6. Add the squared values.
7. Take the square root to return from a squared length to a length.

How to use the interactive

- Start with the Orientation warm-up. Do not calculate yet. Just identify the hypotenuse.
- In the Movement section, choose one example at a time.
- Complete the worked solution from Step 1 to Step 6.
- Use Focus this step when you want the Socratic Prompt and Hint buttons to support a particular step.
- Use Check this step to test one step, or Check all after you have tried the whole route.
- Use the Scratchpad when you want to sketch, copy a step, or make a teaching card.

The method in one line

right angle -> opposite side -> hypotenuse -> square -> add -> square root

Scratchpad reminder

The Scratchpad is optional. It includes Pen, Line, Colour, Width, Eraser, Undo, Redo, Sticky Note and Background. You can use the Line tool to draw a clean hypotenuse or construction line. Erasing your marks should not erase the selected background.

What good evidence looks like

- The right angle is identified.
- The hypotenuse is named correctly.
- The correct letter is used on the left of the equation.
- The square calculations are visible.

- The addition step is visible.
- The square root step is visible.
- The final answer is checked: the hypotenuse must be the longest side.

Confidence check

I can...	Not yet	Nearly	Secure
Find the right angle before calculating.			
Name the hypotenuse as the side opposite the right angle.			
Use a, b or c correctly from the vertex convention.			
Build the worked solution without skipping the square-root step.			
Explain why my answer makes sense.			

Filename pattern

NEO_MATH_PYTH_HYPOTENUSE_FirstName_Date